



AJEENKYA
D Y PATIL UNIVERSITY
THE INNOVATION UNIVERSITY

A
MINI PROJECT REPORT ON
Prediction of Student Performance Based on Study
Hours

FOR

Term Work Examination

***Bachelor's of Computer Application in Artificial Intelligence
and Machine learning(BCA - AIML)***

Year 2024-2025

Ajeenkya DY Patil University, Pune

-Submitted By-

Mr.

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Date: 14 /04 / 2025

CERTIFICATE

This is to certified that **Aniket Sargar**
A student's of **BCA(AIML), Sem-IV, URN No: 2023-B-25082005B**
has Successfully Completed the Dashboard Report On

Prediction of Student Performance Based on Study Hours

As per the requirement of
Ajeenkya DY Patil University, Pune was carried out under my
supervision.

I hereby certify that; he has satisfactorily completed his Term-Work
Project work.

Place: - Pune

Examiner

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Mini Project Report

Project Title :

Prediction of Student Performance Based on Study Hours

Introduction:

In the current educational environment, student performance prediction has become an essential tool for identifying learning trends and providing customized support. This project aims to analyze and predict student performance based on the number of study hours per day. By utilizing historical data and machine learning techniques, we can draw meaningful insights and create a predictive model to estimate expected scores.

Objective:

The primary objective of this project is to analyze the relationship between the number of hours a student studies and their academic performance, and to build a simple machine learning model that can predict a student's score based on their study hours. This project aims to help educators and learners understand how consistent study habits can impact academic success by providing data-driven insights through visualization and prediction techniques.

Methodology & Approach

The project follows a structured methodology, combining data science and machine learning techniques to predict student performance. The approach is divided into several key stages:

1. Data Collection:

- The dataset used in this project contains two primary columns: Hours (number of study hours) and Scores (percentage achieved).
- The data is sourced from a publicly available dataset suitable for supervised learning.

2. Data Preprocessing:

- Data Cleaning: The dataset is checked for missing values and inconsistencies.
- Data Inspection: Basic statistical analysis (mean, median, standard deviation) is performed to understand the data distribution.
- Since the dataset is clean and simple, no complex preprocessing like scaling or encoding is required.

3. Exploratory Data Analysis (EDA):

- Visual Inspection: A scatter plot is used to observe the relationship between Hours and Scores.
- Correlation Analysis: The correlation coefficient is calculated to measure the strength of the relationship.
- The analysis shows a strong positive linear correlation, suggesting that scores tend to increase with more hours studied.

4. Data Splitting:

- The dataset is divided into training and testing subsets (typically 80% training and 20% testing) using `train_test_split` from Scikit-learn.
- This step ensures that the model is trained on one part of the data and evaluated on another, avoiding overfitting.

5. Model Building:

- A Simple Linear Regression model is used, as the problem involves predicting a continuous numerical output based on a single input feature (Hours).
- The model learns the best-fit line that maps input hours to predicted scores by minimizing the error.

6. Model Evaluation:

- After training, predictions are made on the test data.
- Performance is evaluated using:
 - Mean Absolute Error (MAE) – to measure average prediction error.
 - R-squared (R^2) Score – to measure the goodness of fit (how well the regression line explains the data).

7. Data Visualization:

- The regression line is plotted over the original data points to visually confirm how well the model fits.
- This helps stakeholders quickly understand the model's effectiveness.
- An optional residual plot can be used to visualize prediction errors.

8. User Input & Prediction:

- A simple input interface allows users to enter custom study hours.
- The model then predicts the expected score based on the input using the trained regression formula.

Implementation & Code:

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

# Step 1: Load the dataset
data = pd.read_csv("/content/student_scores.csv")

# Step 2: Visualize the data
plt.scatter(data['Hours'], data['Scores'], color='blue')
plt.title('Study Hours vs Scores')
plt.xlabel('Hours Studied')
plt.ylabel('Score (%)')
plt.grid(True)
plt.show()

# Step 3: Prepare the data
X = data[['Hours']] # Input features (2D)
y = data['Scores'] # Target (1D)

# Step 4: Split the data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=0)

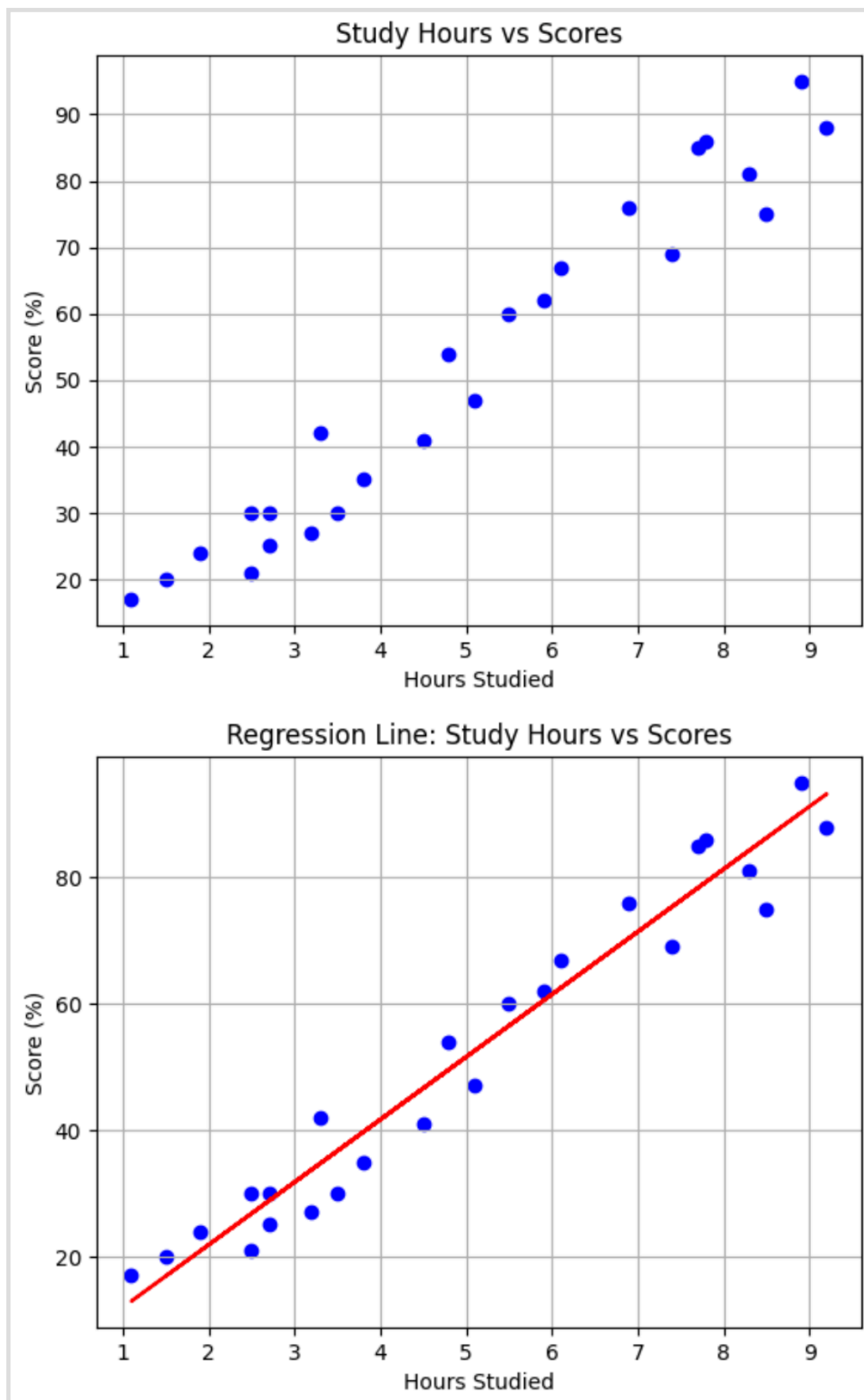
# Step 5: Train the model
model = LinearRegression()
model.fit(X_train, y_train)

# Step 6: Make predictions
predictions = model.predict(X_test)

# Step 7: Plot the regression line
plt.scatter(X, y, color='blue')
plt.plot(X, model.predict(X), color='red') # Regression line
plt.title('Regression Line: Study Hours vs Scores')
plt.xlabel('Hours Studied')
plt.ylabel('Score (%)')
plt.grid(True)
plt.show()

# Step 8: Predict for a custom input
hours = float(input("Enter hours studied: "))
predicted_score = model.predict([[hours]])
print(f"Predicted Score: {predicted_score[0]:.2f}%")
```

Results & Visualisation's



```
Enter hours studied: 12
Predicted Score: 120.95%
/usr/local/lib/python3.11/dist-packages/sklearn/
utils/validation.py:2739: UserWarning: X does not
have valid feature names, but LinearRegression was
fitted with feature names
```

Conclusion & Future Scope:

Conclusion:

This project successfully demonstrates that the number of study hours is a strong predictor of student performance. A simple linear regression model proved effective with high accuracy, making it useful for academic insights.

Future Scope:

- Include more features such as sleep hours, previous scores, or classroom participation.
- Use advanced models like Decision Trees or Random Forests for improved predictions.
- Build a full student dashboard using Streamlit or Flask to visualize predictions interactively. warnings.warn(Visualisation's